Representation of Dictionary Files

Each bilingual dictionary contains the following files and folders:

language1.txt: This file contains the dictionary entries for langauge-1 in the 5 line DSFRecord format.

The DSFRecord Format is modified to accommodate cross-link information. The DSFRecord will be represented as shown in figure 1.

ID	999999
CAT	VERB
CONCEPT	<Concept explanation>
EXAMPLE	<Example>
SYNSET-HINDI	W1/HIN1, W2/HIN3, W3/HIN2

1 -- DSFRecord - Modified

Here, W1, W2 and W3 are the words in the synset. Word W1 is cross-linked with the first word of Hindi (pivot) Synset 999999 by writing "/HIN1" beside W1. (Note: The count starts from 1, not from 0)

ID	591
CAT	VERB
CONCEPT	किसी पर किसी वस्तु आदि से आघात करना
EXAMPLE	"सिपाही चोर को लाठी से मार रहा है / उसने बच्चे को एक चाटा रसीद किया"
SYNSET-HINDI	मारना, पीटना, प्रहार_करना, ठोंकना, ठोकना, पीटाई_करना, धुनना, धुनाई_करना, ताड़ना, प्रताड़ना, रसीद_करना

2 -- DSFRecord for Hindi Synset – 591

ID	591
CAT	VERB
CONCEPT	एखाद्या वस्तूने कोणावर आघात करणे
EXAMPLE	"चोराने मागून मारले."
SYNSET-MARATHI	मारणे/HIN1, प्रहार_करणे/HIN3, घाव_घालणे

3 -- DSFRecord for Marathi Synset – 591

The word can be neglected for cross-linking by simply not providing above information. An example will make this clear. Consider Hindi and Marathi synset with synsetID - 591 as shown in figure 2 and 3.
Here,
मारणे is cross-linked with Hindi word मारना by writing मारणे/HIN1 in Marathi synset.
प्रहार_करणे is cross-linked with Hindi word प्रहार_करना by writing प्रहार_करणे/HIN3 in Marathi synset.
घाव_घालणे is not cross-linked with any Hindi word.

language2.txt: This file contains the dictionary entries for langauge-2 in the 5 line DSFRecord format.

language1 folder: This folder contains 3 types of files namely <pos_tag>.idx, data.<pos_tag> and <pos_tag>-crossLinks.idx. Since the dictionary supports 6 POS Tags (i.e. noun, verb, adverb, adjective, FW and TAM) there will be a total of 18 files in this directory. The contents of these files are as described below:

i. <pos_tag>.idx: This file contains a sorted list of all the words belonging to this <pos_tag> in the dictionary and the corresponding synsets to which each word belongs. The format of each line in this file is as show below:
word<space>***[a space separated list of synset ids to which the word belongs]***

A few entries from the file noun.idx are shown below,

अंकुश 14151

अंकुसी 3279 7879 13664 13665 13666 14153

अंकोर 1958 1960 5065 5841 14158

अंकोरी 1958 1960 3567

अंकौर 1958 1960

ii. data.<pos_tag>: This file contains a sorted list of all the synsets belonging to this <pos_tag> . The format of each line in this file is as show below:
<synset_id>:~:***<pos_tag>***:~:***<concept>***:~:***<example>***:~:***<synset_members>***

A few entries from the file data.noun are shown below,

10001:~:NOUN:~:वे नृत्य या नाच जो गाँवों में नाचे जाते हैं :~:"विद्यालय के वार्षिक समारोह में बच्चों ने तरह-तरह के लोकनृत्य प्रस्तुत किए":~:लोकनृत्य, लोक-नृत्य

10009:~:NOUN:~:लौटने की क्रिया या भाव:~:"श्याम अपने गाँव लौट गया":~:लौट

10011:~:NOUN:~:वकील का काम या पेशा :~:"आजकल राम की वकालत अच्छी चल रही है":~:वकालत

iii. <pos-tag>-crossLinks.idx: This file contains a sorted list of all words of all synsets belonging to this pos-tag. The format of each line in this file is as follows:
<word><synsetId>[\t<langName>-<cross-linkedWord>]*

An entry from the file noun-crossLinks.idx looks as shown below.
अंबुज1628 URD-کمل PAN-ਕਮਲ ORI- ENG- KAN-ನೀಲ_ಕಮಲ MAL-താമര GUJ- TEL-తామర HIN-अंबुज BEN-পদ্ম

<u>language2 folder</u>: The contents of this folder are the same as that of language2 except that the files have data corresponding to language2.

Adding synsets to user_dictionary for a language pair (say, Hindi and Telugu):

1. Go to the languauge-pair directory e.g. hin-tel.
2. Open the file hin.txt. Add a new synset as a five line entry as shown below.

   ID :: 48

   CAT :: ADJECTIVE

   CONCEPT :: बिना नाम का या जिसका कोई नाम न हो

   EXAMPLE :: "रामू ने अनाथ आश्रम से एक अनाम बालक को गोद लिया"

   SYNSET-HINDI :: अनाम

3. Open the file tel.txt. Add the corresponding Telugu synset as a five line entry as shown below.

   ID :: 48

   CAT :: ADJECTIVE

   CONCEPT :: పేరులేని లేదా ఏ పేరూ లేని వ్యక్తి

   EXAMPLE :: "రాము ఒక అనాథ ఆశ్రమం నుండి ఒక అనామక బాలుడిని దత్తత తీసుకున్నాడు

   SYNSET-TELUGU :: అనామక

Please make sure that the new synset-id is unique and does not clash with existing synset ids. Also the hindi and telugu synset-id should be the same.

4. Run the script with the options shown below:

update-dictionary -i $user_dictionary/hin-tel -o $user_dictionary/hin-tel -sHIN -tTEL

5. The script will create the two folders (in this case, hin and tel) if they do not already exist. It will also create 18 files for each langauge.